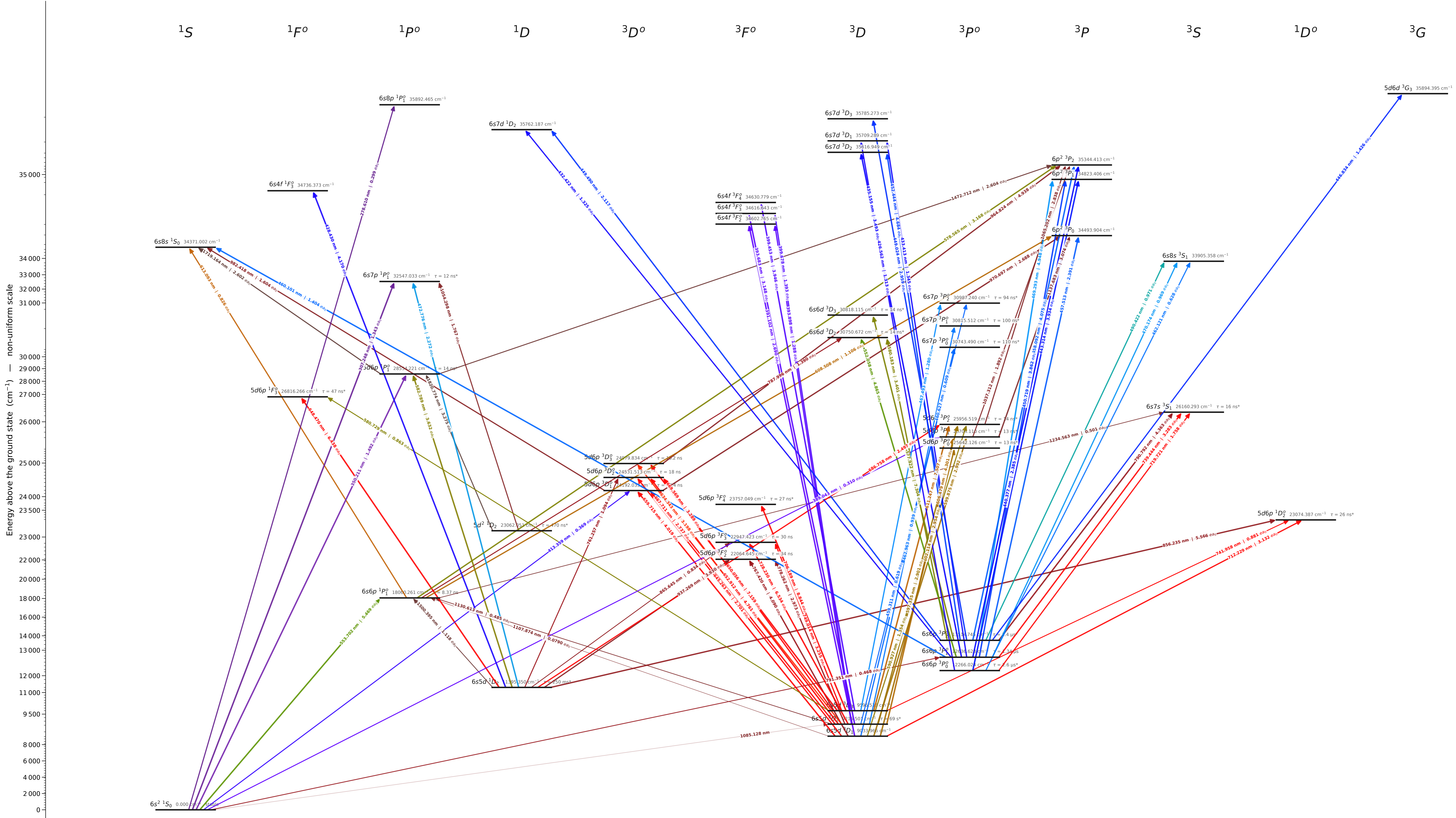
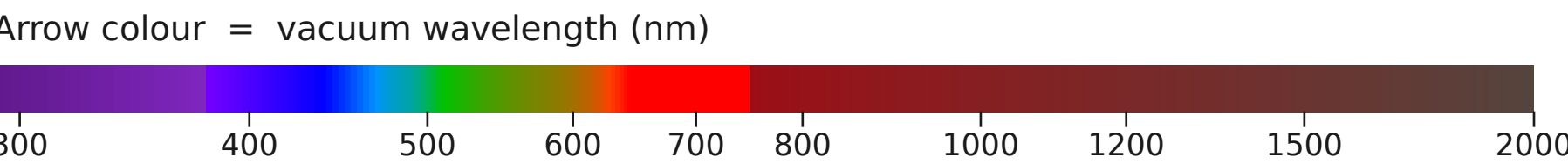


Barium-138

¹³⁸Ba (Ba I), nuclear spin *I* = 0. Energy levels and transitions below 2 μm: 44 levels, 89 lines.

Ba ⁺ ionisation limit 42034.91 cm⁻¹ = 5.21166 eV (237.897 nm)

Arrow label: λ reduced dipole matrix element $|U||er||U|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er U (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
6s ² 1S ₀ – 5d6p ³ D ₂ ^o	413.3592	413.2427	725.258895(5)	0.369	1.300e+06	9.147	2.3	413.36 nm intercombination line: fluorescence detection o...
6s ² 1S ₀ – 6s6p ¹ P ₁ ^o	553.7019	553.5481	541.4330	5.469	1.190e+08	19.01	99.7	main cooling / imaging line
6s5d ³ D ₁ – 5d6p ³ D ₂ ^o	659.7147	659.5326	454.4274	4.019	3.800e+07	9.147	64	659.7 nm repumper
6s5d ³ D ₂ – 5d6p ³ D ₂ ^o	667.7113	667.5270	448.9851	2.737	1.700e+07	9.147	32	667.7 nm repumper
6s ² 1S ₀ – 6s6p ³ P ₁ ^o	791.3506	791.1330	378.836433(3)	0.468	2.990e+05	0.1183	40.2	intercombination line
6s ² 1S ₀ – 6s5d ³ D ₂	1085.1282	1084.8310	276.2738	–	1.454e-02	2.307e-09	100	E2 decay limiting the 3D2 lifetime
6s5d ³ D ₁ – 6s6p ¹ P ₁ ^o	1107.8743	1107.5709	270.6015	0.079	3.100e+03	19.01	0	1107.8 nm decay branch / repumper
6s5d ³ D ₂ – 6s6p ¹ P ₁ ^o	1130.6129	1130.3034	265.1592	0.485	1.100e+05	19.01	0.09	1130.6 nm decay branch / repumper
6s5d ¹ D ₂ – 6s6p ¹ P ₁ ^o	1500.3951	1499.9852	199.8090	1.118	2.500e+05	19.01	0.25	1500.4 nm decay branch of the cooling line / repumper

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
6s ² 1S ₀ – 5d6p ³ D ₂ ^o	413.359	137-138	392.9(9)
6s ² 1S ₀ – 5d6p ³ D ₂ ^o	413.359	136-138	178.1(8)
6s ² 1S ₀ – 5d6p ³ D ₂ ^o	413.359	135-138	401.4(12)
6s ² 1S ₀ – 5d6p ³ D ₁ ^o	413.359	134-138	124.3(13)
6s ² 1S ₀ – 6s6p ¹ P ₁ ^o	553.702	137-138	215.15(16)
6s ² 1S ₀ – 6s6p ¹ P ₁ ^o	553.702	136-138	128.0(4)
6s ² 1S ₀ – 6s6p ¹ P ₁ ^o	553.702	135-138	259.29(17)
6s ² 1S ₀ – 6s6p ¹ P ₁ ^o	553.702	134-138	143.0(5)
6s5d ³ D ₁ – 5d6p ³ D ₂ ^o	659.715	137-138	189.2(10)
6s5d ³ D ₁ – 5d6p ³ D ₂ ^o	659.715	136-138	112.9(10)
6s5d ³ D ₁ – 5d6p ³ D ₁ ^o	659.715	135-138	228.3(10)
6s5d ³ D ₁ – 5d6p ³ D ₁ ^o	659.715	134-138	125.0(20)
6s ² 1S ₀ – 6s6p ³ P ₂ ^o	791.351	137-138	183.4(10)
6s ² 1S ₀ – 6s6p ³ P ₂ ^o	791.351	136-138	109.2(10)
6s ² 1S ₀ – 6s6p ³ P ₂ ^o	791.351	135-138	219.9(10)
6s ² 1S ₀ – 6s6p ³ P ₂ ^o	791.351	134-138	122.3(25)
6s5d ³ D ₁ – 6s6p ¹ P ₁ ^o	1107.874	137-138	-114(4)
6s5d ³ D ₁ – 6s6p ¹ P ₁ ^o	1107.874	136-138	59.3(6)
6s5d ³ D ₁ – 6s6p ¹ P ₁ ^o	1107.874	135-138	-39(4)
6s5d ³ D ₁ – 6s6p ¹ P ₁ ^o	1107.874	134-138	144.1(10)
6s5d ³ D ₁ – 6s6p ¹ P ₁ ^o	1107.874	137-135	-75.3(5)
6s5d ³ D ₂ – 6s6p ¹ P ₁ ^o	1130.613	137-138	-69(3)

Shift = ν(first isotope) – ν(second isotope).